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Comparison of Direct and Cascaded High-Resolution Spatial Downscaling of Wildfire Data

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Abstract

With climate change, there is an increasing need for higher-resolution wildfire data for better monitoring and risk assessment. Geostationary (GEO) satellites offer high temporal resolution but lack the spatial resolution needed for detailed risk assessment, while Low-Earth Orbit (LEO) satellites offer high spatial resolution but low temporal resolution to monitor rapid wildfire spread. This project investigates deep learning approaches for spatial downscaling of wildfire data from GOES imagery, addressing large resolution gap (2km to 30m) alongside severe class imbalance in the fire target, and incorporating physical covariates (topography and land cover). This project compares hybrid attention U-Net (HAUNet) and a simpler convolutional Autoencoder baseline. Two pipelines are compared: direct downscaling (2km GOES to 30m Landsat) and cascaded approach with VIIRS as an intermediate stage (2km to 375m to 30m) to reduce the resolution gap. Across configurations, absolute performance remains modest, reflecting the difficulty of the resolution gap and the limited available training data. Neither architecture shows consistent advantage: HAUNet generally achieves stronger AUPRC, while the Autoencoder achieves better spatial overlap at a fixed decision threshold. Feature ablations show topographic covariates meaningfully contribute to performance.

Declaration of Software & Generative AI Tools

The content and critical analysis presented in this report were written by the student without generative AI.

Generative AI models (Google Gemini 3.6 Anthropic Claude Sonnet 5) were used as developer tools during software implementation. This usage was restricted to:

- **Code Generation & Modularisation:** Drafting code, modularising scripts, and creating visualisation tools.
- **Debugging & Cloud Integration:** Troubleshooting execution errors, optimisation, and assisting with using Google Cloud Services.
- **Implementation Ideation:** Exploring architectural configurations and workflow design possibilities.

Generated codes were manually reviewed, modified, and integrated by the author. References detailing AI usage is in the project repository at `references.md`.

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1 Introduction

In recent years, due to rising popularity in Net-Zero initiatives, carbon investors are increasingly funding carbon reduction and removal projects, such as reforestation (Lambert and Turner, 2024). However, these carbon projects are at risk because wildfires can destroy and release the carbon back into the atmosphere. With climate change, wildfires are becoming increasingly severe and frequent (Jones et al., 2020). This causes economic losses for the insurance industry. For instance, The January 2025 Los Angeles County wildfires set an all-time record, causing $40 billion in insured losses alone from a single event (Dwyer and Nam, 2026). This project, conducted with Artio, aims to address this gap. By providing highly accurate risk assessment models, this research seeks to give better confidence and enable better risk assessment.

High-spatial and high-temporal resolution wildfire data is favourable in risk assessment. However, high-temporal resolution data obtained from geostationary satellites (GEO) have limited spatial resolution, which makes them difficult to use for accurate risk assessment. While low orbit satellites (LEO) offers high-spatial resolution, but low-temporal resolution.

To bridge this gap, existing methods uses Deep Learning (DL) models for downscaling due to their ability in recognising non-linear patterns. However, the resolution gap between GEO and LEO is significant (2km - 30m). This increases the risk of producing physically implausible or unrealistic outputs, as a single low-resolution (LR) pixel can correspond to different high-resolution (HR) details (Z. Wang et al., 2020). In this project, to mitigate this limitation, the model integrates additional physical information, such as HR topography and landcover to help guide the downscaling process. To further improve downscaling, the project aims to investigate whether cascaded downscaling (2km to 375m to 30m) produces more accurate and better results compared to direct downscaling (2km to 30m). By breaking up the downscaling process, the model trains on a smaller resolution gap, aiming to mitigate error caused by the large resolution difference. This project will compare the cascaded and direct pipeline performance of different downscaling architectures, including hybrid attention U-Net.

2 Current State of Remote Sensing Downscaling

Traditional downscaling methods often fail to capture the complex non-linear patterns in remote sensing data. Recently, DL methods have shown to outperform these approaches (Sdraka et al., 2022). CNN and U-Net architectures have been widely used for their ability to capture spatial features. While GAN-based methods have been used where sharpness is prioritised over pixel-wise accuracy. More recently, transformer-based models use long-range attention to capture dependencies beyond a convolution's receptive field (J. Wang et al., 2025), and diffusion-based approaches have showed strong generative ability (Y. Wang et al., 2023).

However, many downscaling architectures are optimised for texture-rich images. They are not well-suited to the distinct structure of active wildfire data where the target is very sparse. In the risk industry, fire mask is preferred for easier analysis. This creates two challenges: severe class imbalance, which biases pixel-wise reconstruction losses and convolutional decoders (Qi et al., 2026); and an ill-posed one-to-many mapping, where a single low-resolution pixel have many possible high-resolution configurations (Z. Wang et al., 2020). Architectures are needed that can preserve fire signal against a dominant background and recover coherent fire-front structure.

Models that downscale across large reoslution gaps with only the coarse fire input have to rely on the learned spatial patterns from the coarse data. While wildfires are influenced by factors such as terrain and landcover (Sullivan, 2017), generic super-resolution models omit this information. Providing HR covariates provides additional context and constraints the model can learn from, which can help mitigate the ill-posed nature of the downscaling problem.

Existing literature is dominated by direct, single-model downscaling across the full resolution gap. While multi-stage or task-decoupled downscaling methods exist (such as Li et al., 2023 that

combines spatial mixing with residual compensation, or decoupling classification and regression H. Wang et al., 2026), no prior work researches cascaded DL downscaling, particularly for wildfire applications. Specifically, no prior work evaluates a multi-stage physical downscaling pipeline that sequentially trains model architectures trained with independent satellite target datasets at intermediate spatial resolutions (e.g. training Stage 1 to map 2km → 375m using VIIRS target grids, and Stage 2 to map 375m → 30m using Landsat/Sentinel target grids). Existing research also has not explored the trade-off between the reduced per-stage resolution gap and the accumulation of error across stages.

To address these gaps, the objectives of this project are:

1. Develop a domain-specific, multi-stage downscaling framework tailored to the extreme class imbalance, large resolution gap, and physical constraints of active wildfires.

2. Compare the effectiveness of a Hybrid Attention U-Net (HAUNet) against a baseline convolutional autoencoder in preserving localised active fire signals.

3. Evaluate the performance and trade-offs of a multi-stage pipeline against direct downscaling, assessing whether cascaded approach improves overall active fire spatial accuracy.

3 Data and Materials

Table 1 shows the data sources used for this project. The specifications for the VIIRS, GOES, and Landsat data are outlined in Table 2.

The GOES-16/17 data were obtained via public Google Cloud Platform (GCP) buckets (NOAA, 2017, 2018). DEM, and ESA WorldCover are obtained via Google Earth Engine (GEE) Python API (Gorelick et al., 2017) and exported directly to Google Cloud Storage (GCS). The Landsat-8 fire masks are derived by de Almeida Pereira et al., 2021 and obtained from the author's public dataset (hosted on Google Drive) and re-uploaded to GCS for pipeline consistency. VIIRS fire detections are obtained from NASA FIRMS archive. Manifests, split assignments and model checkpoints are also stored in GCS.

There are 12,504 and 1,202 total VIIRS and Landsat patches respectively. The assymetry in sample counts is due to different data sources, Landsat from a pre-existing external dataset, VIIRS generated directly from a larger available detection archive. This gives Stage 1 (GOES → VIIRS) of the cascaded pipeline a larger, more reliable training set, since Stage 1 errors propagate directly into Stage 2's input. But the final 30m downscaling is trained on the same Landsat dataset.

The data was processed so that all regions of interest (ROIs) are dynamically reprojected to their localised UTM zone based on centroid coordinates. Detections from the NASA VIIRS S-NPP archive were filtered for "nominal" and "high" confidence points. Detections are clustered into 7680 × 7680 m grid cells within a 15-minute temporal window of peak activity. VIIRS grid cell pixels record the peak Fire Radiative Power (FRP). Due to extreme class imbalance of the target fire datasets (mean fraction of positive target pixels per patch are 0.85% for VIIRS, 0.06% for Landsat), stratified sampling was applied across temporal extent and fire pixel count bins during data splitting. The closest temporal GOES-16/17 scan within a ±60-minute guard window was queried.

GOES-16/17 channels 7 (3.9 μm, shortwave IR) and 14 (11.2 μm, longwave IR) are used as they directly correspond to the spectral bands for fire-detection for their sensitivity to fire radiance. This is similar spectral logic in VIIRS's own active-fire algorithm, which similarly relies on a shortwave (I4, 3.74 μm) and longwave (I5, 11.45 μm) channel pair. (Schmidt et al., 2013)

In addition to the native 3.9 μm (Ch. 7) and 11.2 μm (Ch. 14) GOES brightness temperatures, a Thermal Fire Index (TFI) is computed as additional input channel:

$$TFI = Ch.7 - Ch.14 \tag{1}$$

Explicitly providing the difference reduces the burden on early network layers to discover a known, physically meaningful relationship, aiming to save model capacity and improving convergence.

Table 1: Data Sources and Specifications

Data Source	**Res.**	**Bands Used**	**Role / Context**
Landsat-8 Fire Masks	30 m	Active Fire Mask de Almeida Pereira et al., 2021	Target / ground truth; binary mask (256×256 px)
VIIRS S-NPP	375 m	I4 (3.74 μm, Active Fire)	Intermediate target / cascaded-pipeline input (FRP, NASA FIRMS)
GOES-16/17 (ABI)	2000 m	Ch. 7 (3.9 μm), Ch. 14 (11.2 μm)	Coarse fire input; brightness temperature, high temporal frequency
Copernicus DEM	30 m	DEM	Static covariate; terrain effects on fire spread
ESA WorldCover	10 m	Landcover (10 classes)	Static covariate; land cover map

4 Methods

4.1 HAUNet (Hybrid Attention U-Net)

Among CNN-based approaches, HAUNet (J. Wang et al., 2023) uses hybrid attention to capture both local fine detail and broader contextual structure across multiple scales.

4.1.1 Hybrid Attention Mechanisms

- **Channel Attention Module (CEM):** Computes channel dependencies with $O(C^2)$ complexity. Given feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions followed by 3×3 depthwise convolutions. The channel attention map $\mathbf{A}_c$ is derived as:

$$\mathbf{A}_c = \text{Softmax}\left(\text{ReLU}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\alpha}\right)\right)\mathbf{V} \tag{2}$$

  where α is a learnable scaling parameter.

- **Spatial-Channel Attention (S_CEM):** Parallel spatial attention branch operating along transposed dimensions $\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s \in \mathbb{R}^{B \times N_{head} \times HW \times C}$. The full spatial attention step scales quadratically with spatial resolution as $O((HW)^2)$.

Table 2: Data Pipeline Configuration Specifications

Parameter	Value	Description
VIIRS Geographic Bounds	$[-170.0, -30.0]$	Longitude limits for the Americas
VIIRS Date Range	2020-01-01 – 2020-12-31	Temporal query window for VIIRS detections
Landsat Geographic Bounds	N. & S. America	Locations of interest for GOES detections
Landsat Date Range	Aug–Sep 2020	Temporal window for Landsat detections de Almeida Pereira et al., 2021
Min. Landsat Active Pixels	10	Minimum active fire pixels to retain a patch
GOES Temporal Guard	60 min	Max. gap between target time and GOES scan
GOES Native Scale	2000 m	Native GOES ABI grid spacing
Landsat Grid Scale	30 m	Resolution, Landsat branch
VIIRS Grid Scale	375 m	Resolution, VIIRS active fire detections
Landsat Dimensions	256×256 px	Target patch size
VIIRS Dimensions	20×20 px	Target size (7680/375)
GOES Dimensions	4×4 px	Target size (7680/2000)

4.1.2 Full-Resolution Covariate Pyramid Branch

The original J. Wang et al., 2023 HAUNet applies full spatial attention in S_CEM imposes an $O((HW)^2)$ memory footprint that causes out-of-memory errors on full-resolution inputs. Hence the inputs (256×256) are downsampled by factor $D = 4$ (i.e. $256/4 = 64$) before passing to the model.

The parallel Covariate Pyramid Encoder was introduced as modification to the HAUNet because downsampling covariates alongside the fire input discards the high-resolution detail at the target 30m scale. Processing covariates at full resolution in a separate branch, then progressively fusing into the main decoder path. This branch processes full-resolution covariate features using Windowed Spatial-Channel Attention Blocks (Windowed-S_CEM) (Liu et al., 2021) for computational efficiency.

For a feature map partitioned into non-overlapping local windows of size $P \times P$ ($P = 8$), Windowed-S_CEM computes spatial attention independently within each window. The window is shifted sequentially with shift size $P/2$. This keeps the spatial attention complexity to $O(P^4)$ per window, independent of overall spatial input dimensions.

4.1.3 Multi-Scale Refinement and Feature Fusion

The main encoder-decoder path produces low-resolution feature representations (64×64). The final refinement stage produces the final high-resolution output. At each stage:

1. Low-resolution features are upsampled via bilinear interpolation followed by 3×3 convolutions.

2. Upsampled features are concatenated with matching scale representations extracted from the Covariate Pyramid Encoder.

3. A 3×3 fusion convolution maps concatenated features back to channel dimension.

The final layer of the HAUNet refinement network outputs prediction at the resolution of the full covariate grid (256×256). Operating the decoder at full resolution preserves fine-grained spatial feature representations across skip connections and covariate channels. This is same for Stage 1 (GOES $\rightarrow$ VIIRS) model output for cascaded pipeline. Rather than constraining the

output's resolution to match the coarse VIIRS target grid (20×20), HAUNet outputs high-resolution spatial feature maps. During metric evaluatation of the Stage 1 model, the outputs are pooled to VIIRS resolution to match VIIRS target.

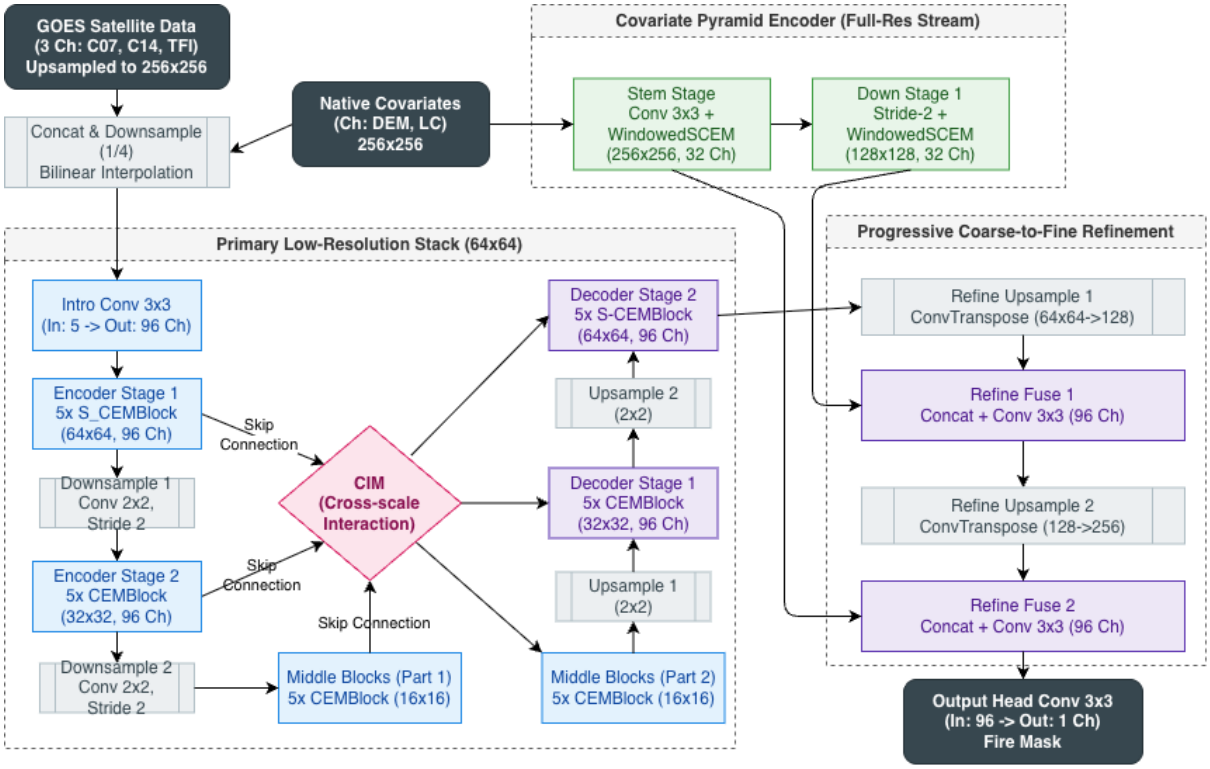


Figure 1: Architecture overview of the wildfire-adapted HAUNet. The full-resolution stream processes native covariates, which are progressively fused via refinement blocks with the low-resolution stack after CIM.

4.1.4 Architecture Selection Rationale

HAUNet was selected as one of the architectures for the following reasons:

1. It uses dual attention: channel attention to capture global information across feature channels and spatial attention to capture spatial relationships. This is useful for wildfire data, where the fire signal is sparse and localised, but the physical context (e.g. terrain, landcover) is important for accurate downscaling.

2. Its CIM used as skip connections. Skip connections used in conventional U-Net concatenate features at matching spatial resolution, but this assume they are semantically comparable. CIM addresses this by explicitly bridging semantic and resolution gaps between encoder outputs at different scales prior to fusion. This helps to better integrate and learn representation at different scales, which is useful for super-resolution tasks like this (downscaling by 66.7×).

3. Both the autoencoder baseline (4.2) and HAUNet share the similar underlying encoder-decoder-with-skip-connections structure. This allows the comparison between the two models to directly isolate the contribution of HAUNet's attention mechanisms and skip-connection design.

4.2 Autoencoder Baseline

The baseline architecture follows the autoencoder proposed by Badhan et al., 2024 to downscale fire data from GOES to VIIRS resolution. Here, it is adapted to this project to concatenate the covariates with the fire data inputs, and outputs are fire mask segmentation rather than continuous BT values. The autoencoder follows a standard U-Net-style structure, full architecture is shown in Figure 2.

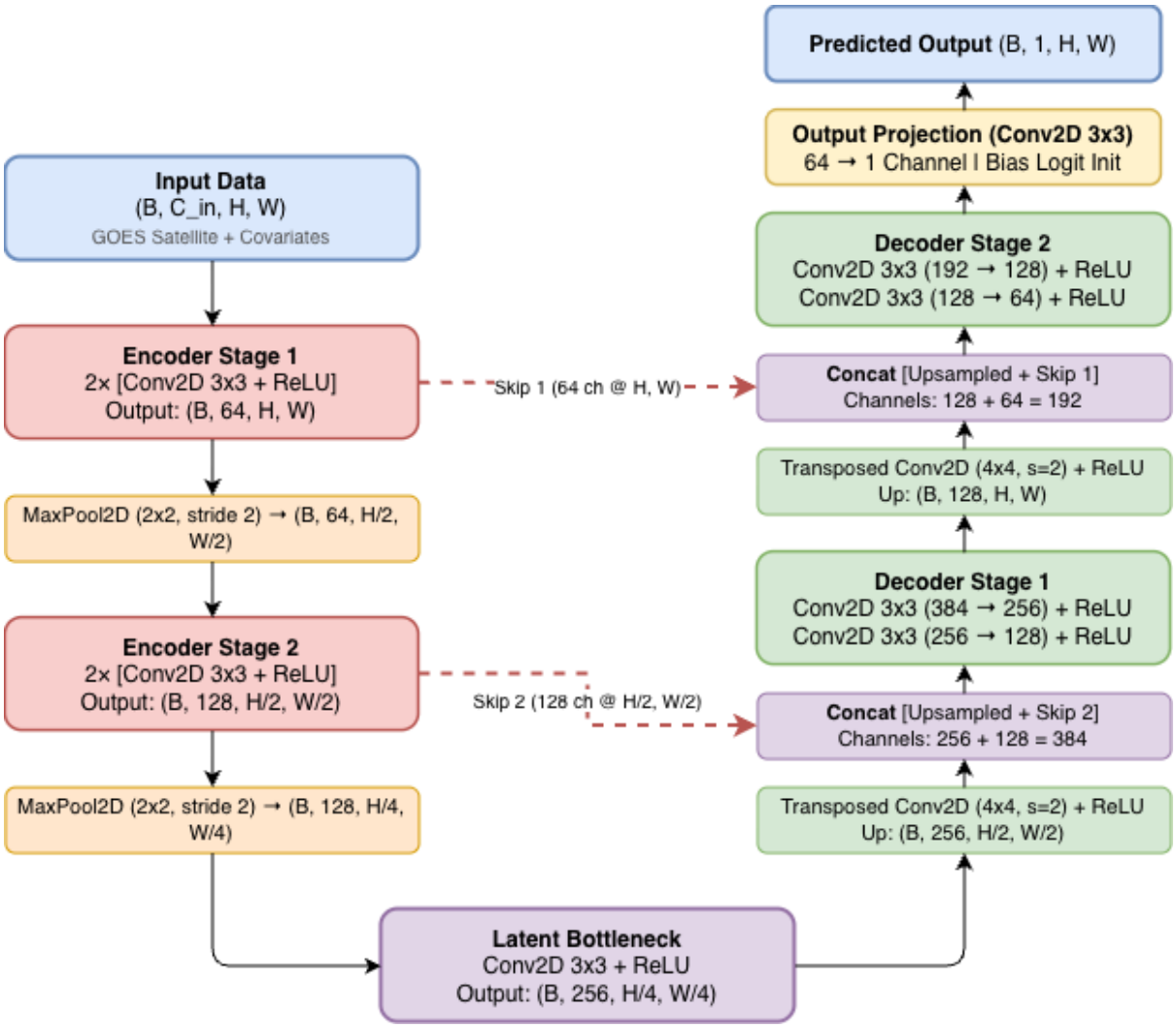


Figure 2: Architecture overview of the autoencoder baseline based on Badhan et al., 2024.

This baseline is trained under identical conditions to HAUNet (same loss, optimiser, data splits, and covariate set) so that any performance gap are attributable to architectural differences. It establishes a lower bound against which HAUNet's improvement can be quantified, and isolating whether performance gains is from attention and full-resolution covariate processing, or would be achievable with a standard convolutional model.

4.3 Loss Functions

4.3.1 Focal Dice Loss

Active fire pixels represent a small fraction of each patch, resulting in extreme class imbalance. To address this, Focal Dice Loss was used. It combines Focal Loss (which down-weights background pixels via $(1 - p_t)^\gamma$) with Dice Loss, which optimises for spatial region overlap:

$$\mathcal{L}_{\text{FD}} = w_f \cdot \mathcal{L}_{\text{focal}} + w_d \cdot \mathcal{L}_{\text{dice}} \tag{3}$$

where w_f and w_d are weights of the losses.

Focal Loss (Lin et al., 2018) applies $(1 - p_t)^\gamma$ to focus training on sparse positives:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{4}$$

where α_t is the balancing factor, p_t is the predicted probability for the ground-truth class, and $p_{i,j} = P(y_{i,j} = 1)$ is the estimated probability for the positive class. For a ground-truth label $y_{i,j} \in \{0, 1\}$:

$$p_t = \begin{cases} p_{i,j} & \text{if } y_{i,j} = 1 \\ 1 - p_{i,j} & \text{if } y_{i,j} = 0 \end{cases}, \quad \alpha_t = \begin{cases} \alpha & \text{if } y_{i,j} = 1 \\ 1 - \alpha & \text{if } y_{i,j} = 0 \end{cases} \tag{5}$$

$\alpha \in [0, 1]$ balances positive and negative pixel contributions, and $\gamma \geq 0$ controls the rate at which pixels are down-weighted.

Dice Loss (Milletari et al., 2016) tries to maximise the spatial overlap:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{i,j} p_{i,j} y_{i,j} + \epsilon}{\sum_{i,j} p_{i,j} + \sum_{i,j} y_{i,j} + \epsilon} \tag{6}$$

where ϵ is a smoothing term preventing division by zero.

The hyperparameters used are: $\alpha = 0.85, \gamma = 1.5, w_f = 1.0, w_d = 1.0$, which was assigned for final model training after tuning with subset of training samples. A higher α was assigned to penalise false negatives more heavily given the extreme class imbalance.

4.4 Training Setup

The dataset is split into training/validation/testing (80/10/10). This split utilises stratified random sampling across four quantile bins based on the active fire pixel count per patch for balanced representation of fire sizes. Weighted random sampling is applied during training to address class imbalance, where sample weights w_i for each patch i were assigned based on the count of positive fire pixels n_i:

$$w_i = \max(n_i, 1) \tag{7}$$

where n_i is extracted from the patch metadata manifests.

Inputs are scaled prior to model ingestion. The GOES and DEM channels undergo Z-score standardisation using mean and standard deviation parameters precomputed exclusively from the training split. DEM covariates were normalised by dividing by 1000 (converting to km). The landcover data is also one-hot encoded into 10 channels, each representing a landcover class.

For the targets, the continuous VIIRS FRP values are binarised, and Landsat targets are already binarised. Depending on the experimental configuration, these are transformed using spatial dilation.

Spatial dilation was applied using a 3×3 max-pooling kernel for the VIIRS branch and a 7×7 for the Landsat branch. Landsat targets are much sparser than the VIIRS targets, so a larger dilation kernel was used to provide a comparably denser training signal.

The Landsat branch has a limited number of samples (1,202 total), and given the extreme class imbalance in fire pixel prevalence, training samples were augmented via independent horizontal and vertical flips (each applied with probability 0.5), applied to the input and target. This increases the effective diversity of the training set. No augmentation was applied to validation or test splits.

All models were trained for 20 epochs using the Adam optimiser with an initial learning rate of $\eta = 1 \times 10^{-4}$, decaying to a minimum learning rate of 1×10^{-6}, and Weight decay

was set to 5×10^{-4} for HAUNet and 1×10^{-4} for the Autoencoder. Batch size of 8 were used to accommodate high-resolution multi-channel tensors within GPU VRAM constraints without downsampling spatial patch dimensions.

4.5 Computational Environment

Model development and training were conducted on a GCP Vertex AI Workbench instance equipped with an NVIDIA L4 GPU, 23.7 GB VRAM, 32 GB RAM, 8 vCPUs. Data generation and preprocessing were run as separate CPU-only jobs querying GCS directly.

4.6 Experimental Design

HAUNet was assessed under the following ablations, in addition to the primary direct-vs-cascaded pipeline comparison:

1. with and without the Covariate Pyramid Encoder.
2. with and without DEM as an input covariate.
3. with and without land cover as an input covariate.
4. with and without target dilation (to assess the effect of target representation on model performance).

In the cascaded pipeline, Stage 1 (GOES to VIIRS) is trained on GOES/VIIRS pairs to predict a VIIRS-resolution (375m) fire mask from the GOES input. Stage 2 (VIIRS to Landsat) is intended to further downscale this to Landsat resolution (30m). However, VIIRS and Landsat overpasses times do not overlap, so matching VIIRS/Landsat pairs are not directly available in the same way GOES/VIIRS pairs are. To address this temporal mismatch, Stage 2 is instead trained using the trained Stage 1 model's predictions on GOES/Landsat dataset as inputs, rather than ground-truth VIIRS observations. For each GOES/Landsat pair, the Stage 1 model produces a predicted VIIRS-resolution fire mask from the temporally matched GOES input, and this prediction is paired with the corresponding Landsat ground truth to train Stage 2. This ensures both stages of the cascaded pipeline operate on inputs available at inference time (i.e. Stage 2 never has access to real VIIRS observations at test time either), avoiding a train/inference mismatch that would occur if Stage 2 were trained on ground-truth VIIRS data unavailable in a genuinely cascaded operational setting.

Dilation was investigated for potentially mitigating the severe class imbalance as it may help the model to learn fire regions more effectively. However, dilation can blur the boundaries of fire regions, potentially leading to less precise predictions. The effectiveness of this approach was evaluated through experiments comparing models trained with and without target dilation. Dilation is performed by applying max pooling to the target masks (kernel size $k = 7$ for Landsat). For this experiment, model training and validation is conducted on dilated targets, but evaluation is performed on the original, undilated targets.

Each experiment (architecture, covariate set, pipeline stage) was defined by YAML configuration files specifying model hyperparameters, data paths, loss weights, and training settings for reproducibility.

In addition to default configuration, the autoencoder was also trained using the same target representation as HAUNet's best-performing configuration (dilated target, Section 5).

4.7 Evaluation Metrics

For the Stage 1 cascaded pipeline, to align the model prediction dimensions (256×256) with the target (20×20), an adaptive spatial max-pooling was applied to the output logits before

Table 3: Summary of experimental configurations evaluated.

Exp.	**Model**	**Pipeline**	**Exp Config.**	**Purpose**
E1	HAUNet	Direct	Full (with pyramid encoder, no dilation)	Primary configuration
E1a	HAUNet	Direct	No pyramid encoder	Ablation: pyramid branch contribution
E1b	HAUNet	Direct	No DEM	Ablation: DEM contribution
E1c	HAUNet	Direct	No land cover	Ablation: land cover contribution
E1d	HAUNet	Direct	Full, dilated target (k=7)	Ablation: target representation (dilated vs. default)
E2.1a	HAUNet	Cascaded (S1)	Full, no dilation	Direct vs. cascaded
E2.1b	HAUNet	Cascaded (S1)	Full, dilated targets (k=3)	Ablation: target representation
E2.2a	HAUNet	Cascaded (S2)	Best config. from E1, E2.1a input	Direct vs. cascaded
E2.2b	HAUNet	Cascaded (S2)	Best config. from E1, E2.1b input	Ablation: target representation
E3	Autoencoder	Direct	Full	Baseline comparison
E3d	Autoencoder	Direct	Full, dilated target	Baseline comparison, matched to E1d
E4	Autoencoder	Cascaded	Full, dilated and undilated target	Baseline, cascaded pipeline

calculating loss and metrics. All reported quantitative metrics including IoU, Boundary F1, AUPRC, and Centroid Distance reflect performance evaluated on the coarse 20×20 target.

- **AUPRC (Area Under the Precision-Recall Curve):** Measures the performance of the model across different thresholds, making it suitable for imbalanced datasets (Davis and Goadrich, 2006):

$$\text{AUPRC} = \sum_k (R_k - R_{k-1}) P_k \tag{8}$$

  where P_k and R_k represent precision and recall evaluated at the k-th threshold operating point. This is selected as the primary metric for evaluation as it is threshold-independent and less sensitive to extreme class imbalance. It captures the trade-off between precision and recall across all thresholds.

- **IOU (Intersection over Union):** Measures the overlap between predicted and ground truth. Given a binary prediction mask $\hat{Y} \in \{0,1\}^{H \times W}$ and binary target mask $Y \in \{0,1\}^{H \times W}$:

$$\text{IoU}(\hat{Y}, Y) = \frac{|\hat{Y} \cap Y|}{|\hat{Y} \cup Y|} = \frac{\sum_{i,j} \hat{Y}_{ij} Y_{ij}}{\sum_{i,j} \left[\hat{Y}_{ij} + Y_{ij} - \hat{Y}_{ij} Y_{ij} \right]} \tag{9}$$

- **Boundary F1:** Restricts precision and recall to fire-perimeter only, obtained by morphological erosion of the predicted and ground truth masks. This isolates edge/shape accuracy from interior-region, which pixel-level F1 can mask:

$$\text{F1}_b = 2 \cdot \frac{\text{Prec}_b \cdot \text{Rec}_b}{\text{Prec}_b + \text{Rec}_b + \epsilon} \tag{10}$$

  where Prec_b and Rec_b are precision and recall computed only over boundary pixels of the predicted and ground truth masks.

- **Precision and Recall:** Evaluates the model's ability to correctly identify fire pixels (precision) and capture all true fire pixels (recall) (Saito and Rehmsmeier, 2015):

$$\text{Precision} = \frac{TP}{TP + FP + \epsilon}, \quad \text{Recall} = \frac{TP}{TP + FN + \epsilon} \tag{11}$$

  Where TP, FP, and FN represent true positive, false positive, and false negative counts across the spatial grid, and $\epsilon = 10^{-8}$ prevents division by zero.

For Boundary F1, IOU, Precision, and Recall, the evaluation is performed on post-processed outputs using Otsu's thresholding to binarise the predicted downscaled masks (Zhang and Hu, 2008). This calculates the variance between the foreground and background classes to determine an optimal threshold t^* that maximises the separation between them. Given class probabilities $\omega_0(t), \omega_1(t)$ and means $\mu_0(t), \mu_1(t)$ for pixels above/below threshold t, the variance is:

$$\sigma_B^2(t) = \omega_0(t)\, \omega_1(t)\, \left(\mu_0(t) - \mu_1(t)\right)^2, \qquad t^* = \arg\max_t \ \sigma_B^2(t) \tag{12}$$

Otsu threshold is calculated and applied per patch. Otsu is used over a fixed threshold (e.g. 0.5) because the extreme class imbalance means that the model may predict low probabilities for the positive class, and a fixed threshold could lead to poor separation.

To find uncertainty of metrics given the limited test-set size, 95% confidence intervals (CI) were computed on AUPRC via percentile bootstrap resampling at the patch level (1,000 resamples with replacement). To assess whether differences between configurations were statistically meaningful, paired bootstrap tests were performed for selected comparisons: the same resampled patches were used to compute AUPRC for both configurations in each resample, giving a

distribution of paired differences from which a 95% CI on the difference, and the proportion of resamples favouring one configuration, were derived. Both procedures reflect sampling uncertainty arising from the finite test set (121 patches) rather than variability across training runs or random seeds.

Due to the extreme rarity of the positive class, absolute metric values are difficult to interpret without a reference point. The following baselines were used:

1. an all-background predictor, assigning zero probability to every pixel.
2. a constant-prevalence predictor, assigning every pixel the measured test-set fire-pixel prevalence.

2. is the theoretically expected value of AUPRC under a classifier with no discriminative ability. Model performance is reported alongside both baselines.

While target dilation were used for the training target to mitigate class imbalance during optimisation, all evaluation metrics were computed against the original, undilated binary target masks. This ensures reported performance reflects the model's ability to predict at native target resolution.

The best model checkpoint for final evaluation was selected based on the highest validation AUPRC. AUPRC evaluates across range of classification thresholds, making it threshold-independent and less sensitive to the extreme class imbalance.

5 Results

The evaluation results for the direct and cascaded pipelines are summarised in Table 4. All metrics in this study are aggregated (per-sample, then averaged across the test set). Example predictions for the best Autoencoer and HAUNet configuration are shown in Figure 4 and 5 respectively. Figure 3 shows the Precision-Recall (PR) curves across representative models compared to the random baseline. Table 5 summarises the best-performing configuration within each architecture/pipeline group

6 Discussions

6.1 Overall Performance

Across configurations, absolute performance is generally low (most < 0.02 AUPRC and IOU). But all exceed both the All-Zero and Prevalence-Rate baselines, meaning the models are learning the fire signal rather than predicting background (relative to prevalence, best direct configuration (E1d) achieves an AUPRC roughly 30 times this baseline, the best cascaded configuration (E2.2a) roughly 23 times, and the Autoencoders (E3, E4.2b) roughly 20 and 17 times respectively).

Direct numerical comparison with similar work in Badhan et al., 2024 is not meaningful given differences in target resolution (2km $\rightarrow$ 375m single-stage vs. this study's 2km $\rightarrow$ 30m direct and cascaded mapping), prediction target (continuous BT regression vs. binary fire segmentation), and evaluation method. However, their IOU values (0.03-0.24) shows that even a single smaller resolution jump, less severe class imbalance (they use 20% fire-coverage threshold, and smaller image 128×128), and more target dataset (their $5,869$ samples vs. $1,202$ Landsat target samples in this study) still results in modest overlap. This suggests that the extreme resolution gap and class imbalance in this study make the low absolute performance unsurprising.

6.2 Cascaded vs. Direct Pipeline

For HAUNet, Stage 2 (E2.2a: IoU 0.0072, Precision 0.0074, AUPRC 0.0142 Boundary F1 0.0071, E2.2b: IOU 0.0089, Precision 0.0095, Boundary F1 0.0104) outperforms the best direct configu-

Table 4: Model performance evaluation across feature ablations vs. fixed baselines. Precision, Recall, Boundary F1 are computed on Otsu-thresholded predictions. Bold indicates the best value for that metric among the shown ablations of the same architecture and target

Exp.	AUPRC	IoU	Precision	Recall	Boundary F1
Direct (GOES->Landsat) - HAUNet					
E1: Full, No Target Dilation	0.0103	0.0021	0.0021	0.3403	0.0030
E1a: No Cov. Pyramid Enc.	0.0098	0.0043	0.0043	0.4605	0.0039
E1b: No DEM	0.0050	0.0020	0.0020	0.3100	0.0033
E1c: No Land Cover	0.0054	0.0026	0.0026	**0.5051**	0.0020
E1d: Dilated Target	**0.0188**	**0.0044**	**0.0044**	0.3490	**0.0041**
Cascaded (GOES->VIIRS->Landsat) - HAUNet					
E2.1a: Stage 1 (no dilation)	**0.1458**	**0.0651**	**0.0721**	0.3582	**0.0956**
E2.1b: Stage 1 (dilated)	0.1408	0.0505	0.0516	**0.6145**	0.0563
E2.2a: Stage 2 (E2.1a input, dilated)	**0.0142**	0.0072	0.0074	0.2065	0.0071
E2.2b: Stage 2 (E2.1b input dilated)	0.0134	**0.0089**	**0.0095**	**0.2435**	**0.0104**
Direct (GOES->Landsat) - Autoencoder					
E3: Full, No Target Dilation	**0.0120**	**0.0064**	**0.0077**	0.2592	**0.0050**
E3d: Dilated Target	0.0098	0.0057	0.0058	**0.5908**	0.0045
Cascaded pipeline - Autoencoder					
E4.1: Stage 1 (no dilation)	0.0944	0.0327	0.0367	0.3966	0.0500
E4.2a: Stage 2 (no dilation)	0.0106	**0.0065**	**0.0080**	0.2761	0.0037
E4.2b: Stage 2 (dilated targets)	**0.0106**	0.0056	0.0058	**0.4565**	**0.0040**
Baseline: All-Zero	0.0006	0.0000	0.0000	0.0000	N/A
Baseline: Prevalence-Rate	0.0006	0.0000	0.0000	0.0000	N/A

Table 5: Best-performing configuration (by AUPRC) within each architecture/pipeline group, for direct cross-group comparison. AUPRC values shown with 95% bootstrap confidence intervals.

Group	AUPRC	IoU	Precision	Recall	Boundary F1
Direct - HAUNet (E1d: Dilated Target)	**0.0188 [0.0084, 0.0336]**	0.0044	0.0044	0.3490	0.0041
Cascaded - HAUNet (E2.2a: Stage 2 (E2.1a input, dilated))	0.0142 [0.0071, 0.0237]	**0.0072**	0.0074	0.2065	**0.0071**
Direct - Autoencoder (E3: Full, No Target Dilation)	0.0120 [0.0044, 0.0243]	0.0064	**0.0077**	0.2592	0.0050
Cascaded - Autoencoder (E4.2b: Stage 2 (dilated targets))	0.0106 [0.0044, 0.0184]	0.0056	0.0058	**0.4565**	0.0040

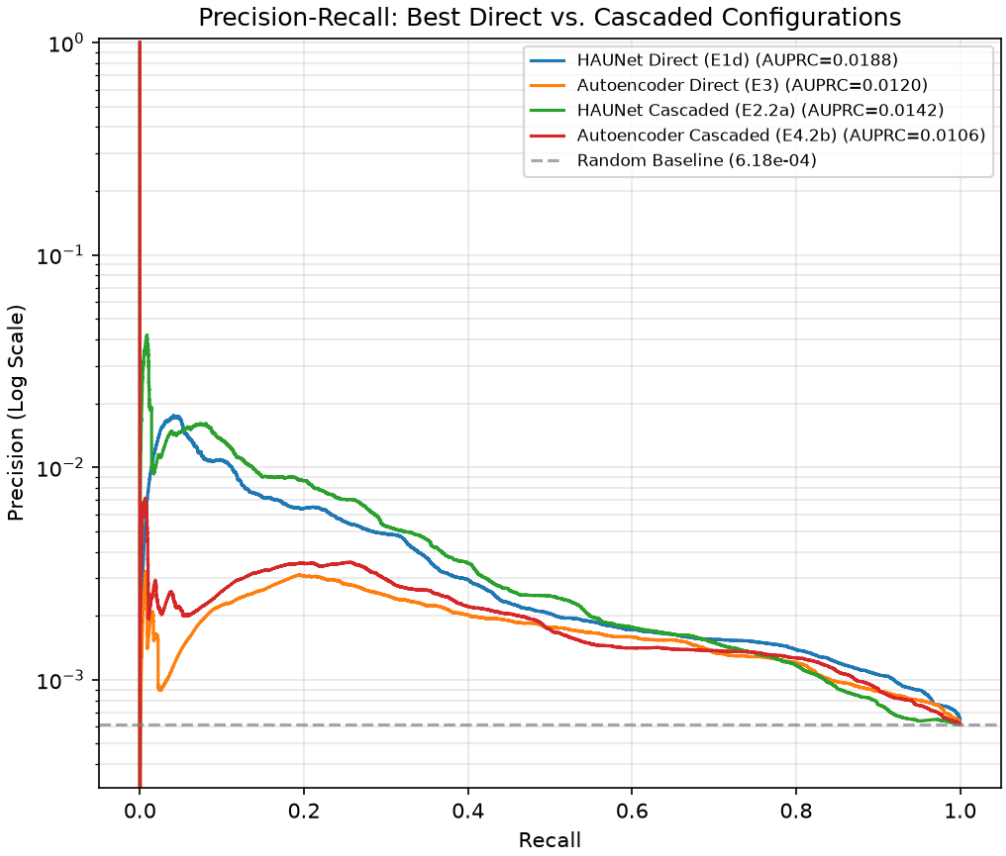


Figure 3: Precision-Recall curves for best Direct and Cascaded configurations (E1d, E2.2a, E3, E4.2b) against the random baseline (Prevalence-Rate = 6.18×10^{-4}). Curves evaluate raw probability outputs across all decision thresholds.

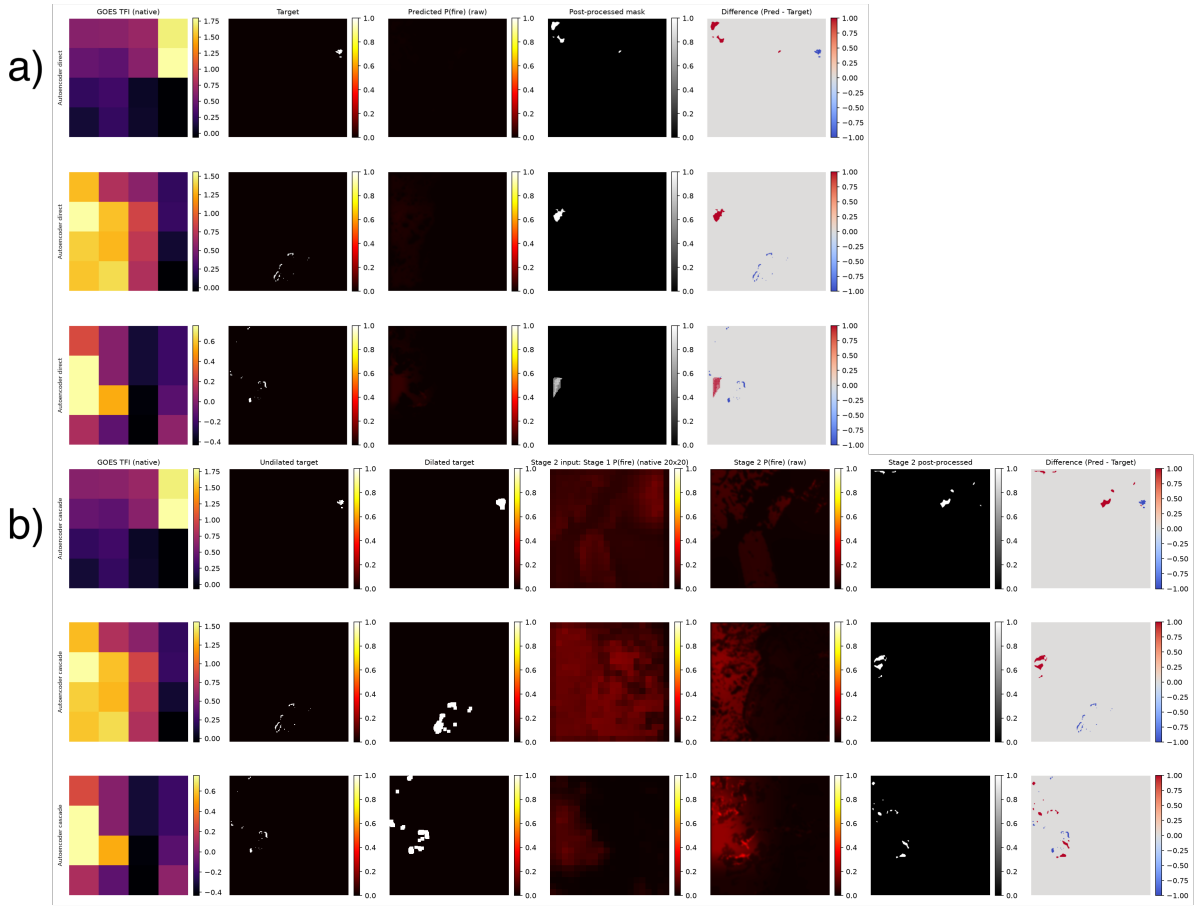


Figure 4: Autoencoder direct E3 (a) and cascaded E4.2b (b) prediction examples. From left to right: a) GOES TFI, undilated target, raw predictions, postprocessed predictions, difference. b) input GOES TFI, undilated target, dilated target, Stage 1 prediction, Stage 2 raw prediction, postprocessed prediction, difference.

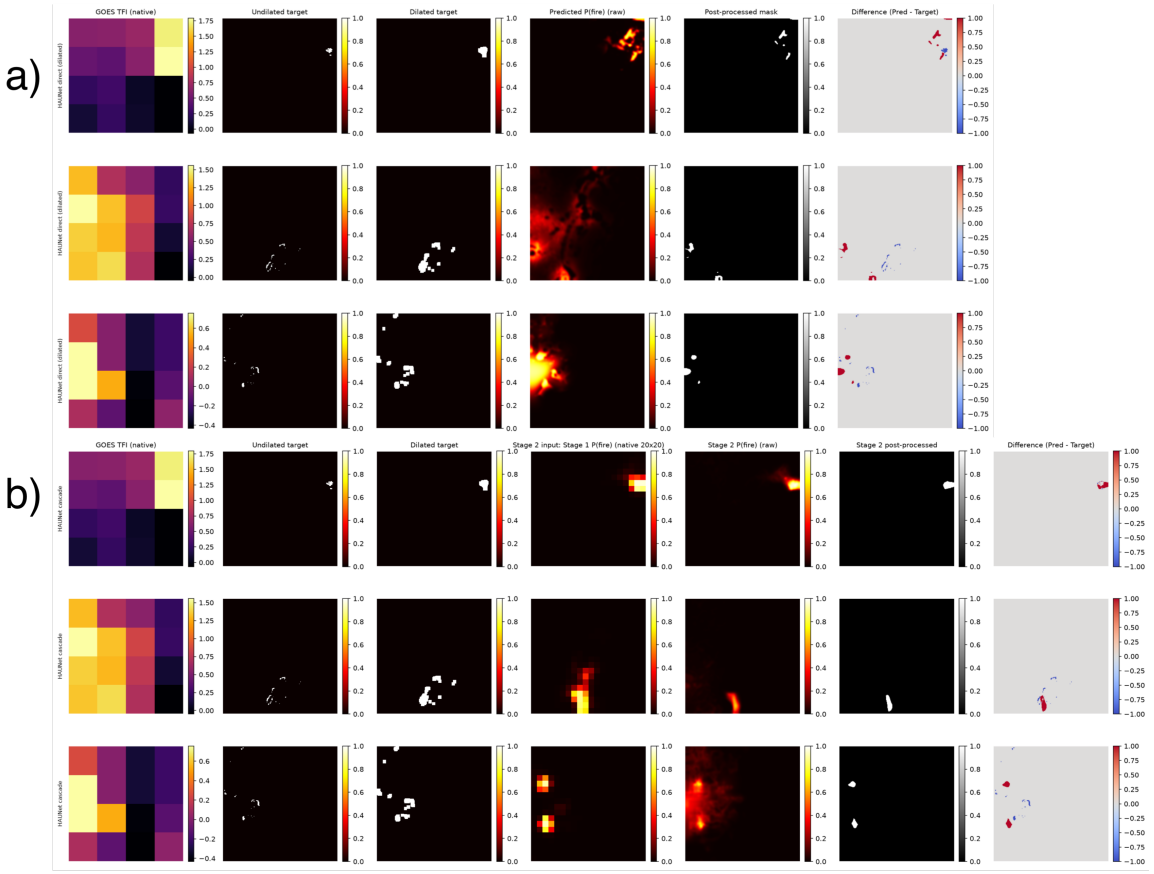


Figure 5: HAUNet direct E1d (a) and cascaded E2.2a (b) prediction examples. From left to right: a) GOES TFI, undilated target, dilated target, raw predictions, postprocessed predictions, difference. b) input GOES TFI, undilated target, dilated target, Stage 1 prediction, Stage 2 raw prediction, postprocessed prediction, difference.

ration (E1d: IoU 0.0044, Precision 0.0044, AUPRC 0.0188, Boundary F1 0.0041) on all metrics except AUPRC and Recall. This suggests the intermediate stage may help the model produce spatially better localised predictions (reflected in overlap-based metrics) but not necessarily improve confidence ranking. Stage 1 metrics are much higher than either Stage 2 or direct-pipeline (e.g. E2.1a AUPRC 0.1458), but this is due to Stage 1 metric being evaluated at VIIRS target and resolution (20×20). So metrics are not directly comparable.

A paired bootstrap test on AUPRC (E2.2a vs. E1d, matched by patch, 1,000 resamples) gives a mean difference of -0.0042 with 95% CI $[-0.0192, 0.0086]$, and E2.2a exceeds E1d in only 30.2% of resamples. While not significant, it is consistent with the direct configuration retaining an AUPRC advantage, even as the cascaded configuration improves overlap-based metrics. Reinforcing that these two metric families capture different aspects of prediction quality.

For the Autoencoder, there is similar pattern. The direct configuration achieves highest AUPRC (E3: 0.0120), while cascaded Stage 2 achieve only slightly better IoU and Precision (E4.2a: IoU 0.0065, Precision 0.0080 vs. E3: IoU 0.0064, Precision 0.0077) and higher Recall (E4.2b: 0.4565 vs. E3: 0.2592). Errors from Stage 1 prediction propagate into Stage 2 training inputs, introducing noise that lowers confidence quality (AUPRC).

A paired bootstrap test on AUPRC (E4.2b vs. E3, matched by patch) gives a mean difference of -0.0013 with 95% CI $[-0.0082, 0.0052]$, and E4.2b exceeds E3 in only 32.3% of resamples, indicating a weaker and less consistent AUPRC gap for the Autoencoder than observed for HAUNet, though neither reaches statistical significance at this sample size.

The divergence between AUPRC and the overlap-based metrics (IoU, Precision, Boundary F1) are seen across comparisons. Because AUPRC is computed over continuous predictions and reflects how well the model ranks pixels by confidence across all thresholds. IoU, Precision, and Boundary F1, are computed only after Otsu thresholding. They depend on the binary mask produced only at one particular threshold. A model can improve performance at specific Otsu threshold (e.g. producing a more spatially coherent, better-localised binary region) without improving its overall confidence across all thresholds. This is a possible explanation for why the cascaded stage, and dilated inputs, improve overlap-based metrics for HAUNet while not improving AUPRC.

6.3 HAUNet vs. Autoencoder Architecture Comparison

Comparing best Stage 2 configurations, HAUNet achieves higher AUPRC point estimates than the Autoencoder in both pipelines:

- **Cascaded Pipeline:** HAUNet leads autoencoder on AUPRC (E2.2a: 0.0142 vs. E4.2a / E4.2b: 0.0106), IoU (E2.2b: 0.0089 vs. E4.2a: 0.0065), Precision (0.0095 vs. 0.008) and Boundary F1 (0.0104 vs. 0.004).
- **Direct Pipeline:** HAUNet leads Autoencoder on AUPRC (E1d: 0.0188 vs. E3: 0.0120), Recall (0.3490 vs. 0.2592). While direct Autoencoder achieves higher IoU (0.0064 vs. 0.0044) and Precision (0.0077 vs. 0.0044).

Paired bootstrap tests on AUPRC support this pattern: HAUNet exceeds the Autoencoder in 84.2% of resamples in the direct pipeline (E1d vs. E3, mean diff +0.0069, 95% CI $[-0.0057, 0.0222]$) and 87.1% of resamples in the cascaded pipeline (E2.2a vs. E4.2b, mean diff +0.0036, 95% CI $[-0.0028, 0.0110]$). Neither interval excludes zero at the conventional 95% threshold given the limited test-set size, but the consistency of this advantage across both pipelines and across roughly six in seven resamples suggests a genuine, if not conclusively significant, architectural difference favouring HAUNet.

Figures 4 and 5 confirm this behavior: Autoencoder predictions produce broader, low-confidence regions, likely due to their convolutional nature. After Otsu thresholding, these wider predictions cover larger ground areas, yielding higher point precision/IoU at the cost of

spatial localisation. HAUNet produces sharper predictions with higher overall confidence ranking (PR curves in 3). Furthermore, HAUNet Stage 1 provides higher-quality inputs to Stage 2 (E2.1a AUPRC 0.1458 vs. E4.1 AUPRC 0.0944) meaning less severe error propagation in the cascaded pipeline. Suggesting the results differ by Stage 2 architecture's own suitability for the task, and differences in the quality of Stage 1 predictions. Investigating these effects require cross-architecture ablation (e.g. evaluating an Autoencoder Stage 2 trained on HAUNet's Stage 1 predictions, vice versa), which is left for future work.

6.4 Over-Prediction, Thresholding, and Practical Implications

Across most configuration, Recall exceeds Precision (e.g. E1c: Recall 0.5051 vs. Precision 0.0026; E1d: Recall 0.3487 vs. Precision 0.0044). This indicates the models are over-predicting fire pixels. However, in insurance, over-predicting may be preferable to under-predicting, as false negatives could have more severe consequences than false positives. It also raises whether Otsu is well-suited to thresholding predictions, as it assumes a roughly bimodal intensity distribution. A fixed threshold or one derived from the validation PR curve may be worth comparing in future work.

To quantify over-prediction, area inflation ratio was computed as predicted fire area divided by true fire area. Median inflation ranged from 39× (E2.2a) to 457× (E4.2b), amplified by the extreme rarity of the positive class (Landsat fire pixels comprise only 0.06% of a patch on average).

For Artio's carbon-investment risk assessment use case, missed fire risk is generally more costly than overestimated risk: the current models are better suited to coarse regional estimates than precise predictions, and any operational use of predicted area would need calibration against this inflation factor. Notably, E2.2a shows lower inflation (39×) than the other configurations, suggesting the cascaded HAUNet may produce more spatially concentrated predictions. Though given the overlapping CIs, this should be treated as observation rather than a confirmed architectural advantage.

6.5 Effect of Target Dilation

Dilating the target mask improves AUPRC and IoU for HAUNet's direct configuration (E1d vs. E1: AUPRC 0.0188 vs 0.0103, IoU 0.0044 vs 0.0021) but not for the Autoencoder (E3d vs. E3: AUPRC 0.0098 vs 0.0120, IoU 0.0057 vs 0.0064, both lower with dilation). This may be because dilation allows for smoothing/tolerance for models like HAUNet that uses attention mechanisms to capture fine-grained information, while Autoencoder may benefit less from this beauce their convolutional nature are prone to smoothing and wider predictions. This is consistent with the higher recall.

6.6 Feature Ablations

Out of direct HAUNet ablations, removing DEM (E1b) produces weakest performance (AUPRC 0.0050, IoU 0.0020) compared to full model (E1: AUPRC 0.0103, IoU 0.0021). This suggests topography is important and greatly contributes to the model's ability to predict fire fronts. Removing land cover (E1c) increases Recall (0.5051, the highest in this group) but lower Boundary F1 (0.0020, the lowest in the group), suggesting that land cover helps produce more spatially precise predictions. Removing the covariate pyramid encoder (E1a) gives intermediate results across most metrics, suggesting it contributes some but not as much as DEM.

7 Conclusions

This project investigated DL pipeline for downscaling low-resolution, high-temporal-frequency GOES imagery into Landsat-resolution wildfire predictions, evaluating both direct and cascaded mapping across HAUNet and Autoencoder baseline. The work was conducted in the context of Artio's carbon-investment risk assessment use case.

Across configurations, absolute performance remained modest (AUPRC and IoU generally below 0.02), showing the difficulty of an extreme resolution gap (2km to 30m), severe class imbalance in the fire target, and small training dataset (1,202 Landsat patches). All trained configurations exceeded the All-Zero and Prevalence-Rate baselines, confirming that the models learn genuine patterns.

No architecture showed a uniform advantage over others. HAUNet consistently achieved higher AUPRC than the Autoencoder in both direct and cascaded mapping, while the Autoencoder generally achieved better overlap-based metrics (IoU, Precision, Boundary F1) in direct mapping. But in cascaded pipeline, HAUNet's Stage 2 outperformed the Autoencoder's across all metrics, though this comparison is likely due to each architecture's Stage 2 being trained on its own Stage 1's predictions. The benefit of cascading over direct mapping was similarly metric-dependent rather than uniform, with cascading generally improving overlap-based metrics for HAUNet at some cost to AUPRC, and showing a smaller, less consistent effect for the Autoencoder. Target dilation improved HAUNet's direct-mapping performance but not the Autoencoder's, suggesting the benefit of label tolerance is architecture-dependent. Feature ablations indicated that DEM contributes meaningfully to HAUNet's direct-mapping performance.

These findings suggest that, with the current data and resolution gap, architectural complexity (HAUNet's attention mechanisms) does not give uniform advantage over a simpler model, and that pipeline design choices (cascaded vs. direct, dilation) interact with architecture in ways that are not yet fully understood. Several possible explanations offered, including the relationship between confidence ranking and thresholded spatial overlap, and the role of Stage 1 prediction quality in shaping Stage 2 outcomes. But these are not verified, and are possible future investigations.

Future work should explore the integration of temporal information to capture the dynamic evolution of wildfires and/or predict burn areas. Another could include integration of climate data, and/or predicting fire intensity as well. Additionally, further research could investigate the use of generative and transformer-based models to enhance downscaled outputs and improve the model's ability to generalise across different geographic regions and fire regimes. The project could be extended to experiment with different architectures at each stage of the cascaded pipeline, or explore multi-stage cascades with more than two stages to further reduce the resolution gap at each step. Increasing the training dataset, and conducting a cross-architecture ablation isolating Stage 2 performance from Stage 1 prediction quality, would help clarify questions raised above.

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